

ENG 401: Interdisciplinary Engineering Design Project – Research & Report Guide

1. Introduction

- ENG 401 is not only about building a product, it's about teamwork. Research, simulations, experiments, and clear reporting are just as essential as the final prototype.
- The power of interdisciplinarity: You can merge insights from mechanical, electrical-electronics, computer, materials, geomatics, civil, biomedical, mechatronics, petroleum engineering, and engineering sciences. Working across fields leads to smarter and more innovative solutions.
- Why reports matter: Your report isn't just paperwork. It's evidence of your thinking, process, and achievements. You may consider it as a storybook of your design journey, showing how your ideas grew, changed, and delivered results.

Tips:

- Have a clear vision for your project from beginning to end.
- Construct your report with evidence-based arguments.

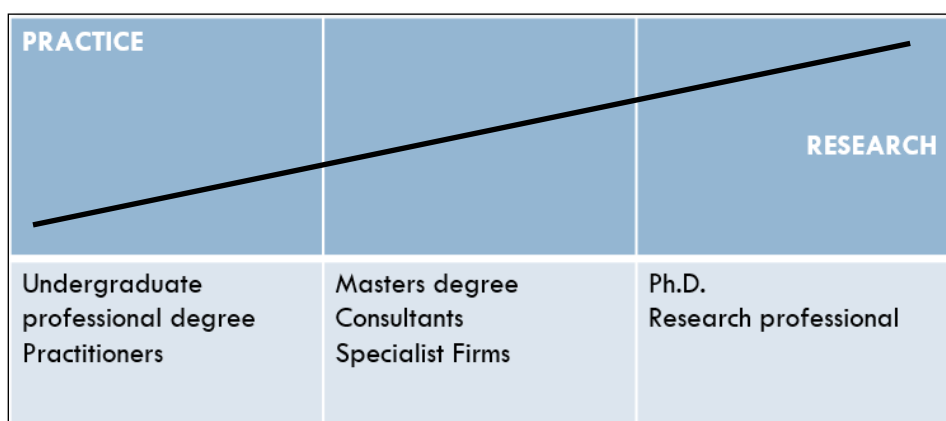
2. Research Methods

What are research methods?

Research methods are the **tools and strategies** to investigate a problem in a structured way. They focus on **how** you conduct research rather than only **what** you study.

In summary, research methods are your **roadmap** for turning a vague idea into reliable, evidence-based knowledge.

Research versus practice



Source: adapted from Groat and Wang, 2002, p. 5

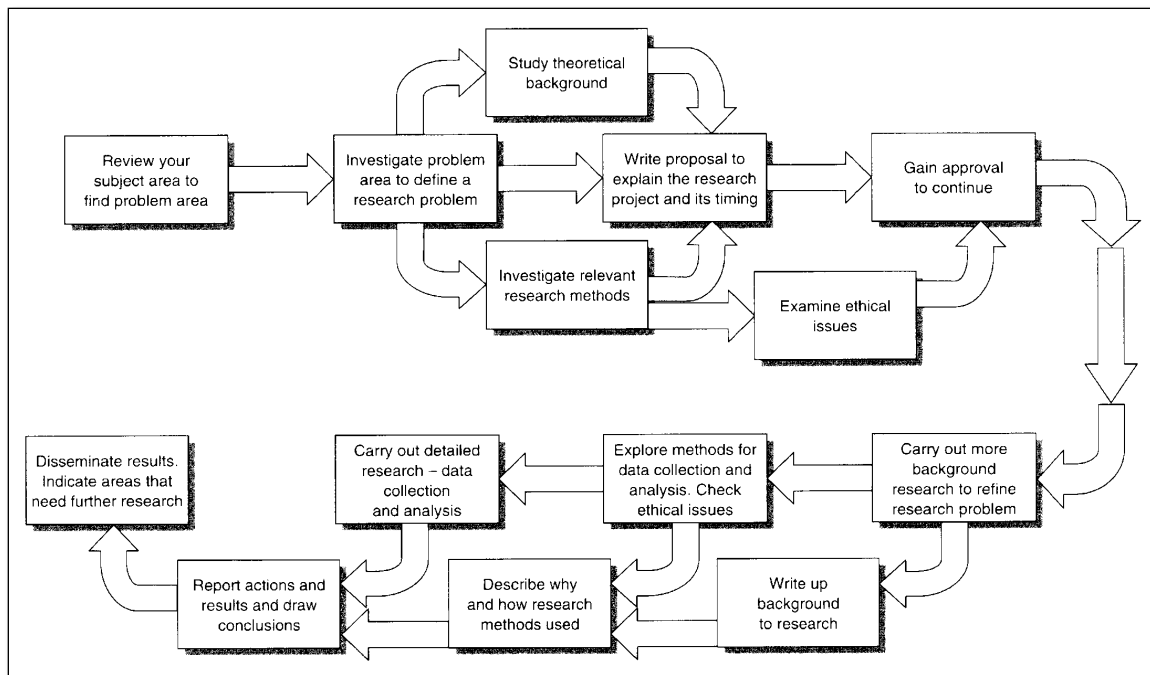


Fig. 1. Typical research process.

2.1 Identifying a Research Problem

When starting your project, the first step is to **define the problem clearly and concisely**.

- **Pick a significant problem** – avoid trivial or repeated work.
- **Express your research intentions** in different but complementary ways:
 - **Research Question:** The main question that define the scope of your study. **Sub-questions** break the main question into smaller, manageable parts.
 - **Aims & Objectives:** State what you want to achieve.
 - **Hypothesis:** A prediction of the expected outcome, supported by reasoning or prior evidence.

Think of it like this: *Your research question asks the “What?”, your objectives explain the “Why?”, and your hypothesis predicts the “What if?”.*

2.2 Literature Review

The literature review is more than a summary, it’s about **understanding what is already known, identifying gaps, and deciding where your work fits in**.

- **Purpose:**
 - Build a foundation of existing knowledge
 - Identify what is missing or uncertain (What is unknown?)
 - Evaluate the reliability of different sources
- **Where to look:**
 - Books and academic journals
 - Online databases and patents
 - Reputable websites and Google Scholar

Tips: When reviewing literature, make a summary by noting:

- Key findings
- Relevance to your project
- Limitations

!A good literature review is not just “what others said”, it’s your **critical evaluation** of the field, showing where your project can contribute something new.

2.3 Determining the Types of Research

Research can be **qualitative** or **quantitative**, sometimes even a mix.

2.3.1 Qualitative

- Focus: **Opinions, behaviors, habits, or human judgments**
- Example: Interviewing users about how they experience wind flow near buildings.

2.3.2 Quantitative

- Focus: **Measurable data, numbers, and variables**
- Two main forms in engineering:

a) Experimental Research

- Core idea: *Change one thing (**independent variable**) and see what happens to another (**dependent variable**).*
- Features: High control over conditions allows you to explore **cause & effect**.
- Stages:
 1. Confirm the problem is suitable for experiments.
 2. Formulate a clear **hypothesis**.
 3. Identify, measure, and control variables.
 4. Design the experiment and run pre-tests.
 5. Collect, analyze, and interpret results.



Fig. 2 View from a wind tunnel experiment.

b) Simulation / Modeling

- Core idea: *Build a simplified version of a system and test scenarios virtually.*
- Tools: ANSYS, STAR CCM+, MATLAB, SolidWorks, COMSOL.
- Key step: Validate models by comparing them with **real-world data** or known benchmarks.

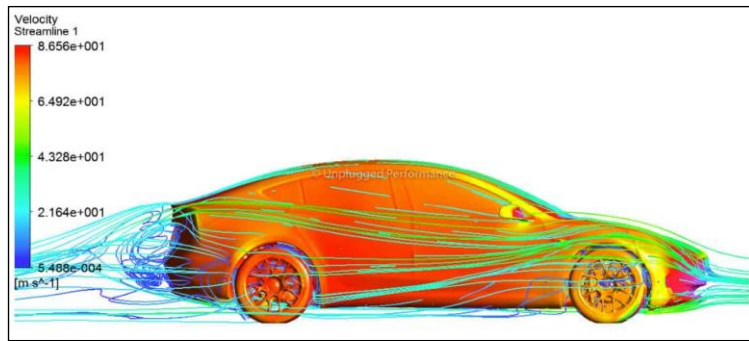


Fig. 3. CFD simulation of building airflow.

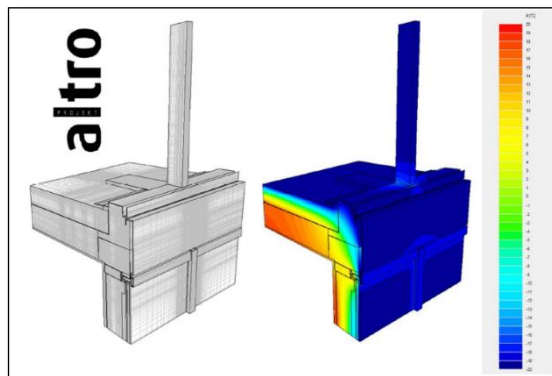


Fig. 4. Thermal simulation of a building facade.

Rule of thumb: Experiment is testing in the real world. Simulation is testing in the virtual world.

Experiment vs Simulation

Aspect	Experiment (Real-World)	Simulation (Virtual)
Time	Can be slow (setup, repetitions)	Faster once model is built
Control	Limited, real conditions may vary	High parameters can be adjusted easily
Safety	Risky in hazardous setups (e.g., high pressure, chemicals)	Safe, no physical danger
Accuracy	High realism, but prone to noise/errors	Depends on model assumptions & validation
Flexibility	Hard to test extreme cases	Easy to run “what if” scenarios

3. Report Writing

Basic Structure:

- **Title Page:** Project title, team members, course, date.

- **Abstract:** A short (150–250 words) summary of your project: problem, method, key results, conclusion.
- **Contents:** Table of contents with page numbers.
- **Introduction:** What problem are you solving? Why does it matter?
- **Methodology:** How did you approach the problem (experiments, simulations, models)?
- **Results:** What did you find? Present data clearly with tables/figures.
- **Discussion:** What do the results mean? Compare to your objectives and literature.
- **Conclusion:** What did you achieve? What are the limitations? Future work?
- **References:** Full details of all sources cited in text.
- **Appendices:** Supporting material (extra data, code, drawings).

Tips for Effective Writing:

- Keep the language **clear and concise**; define technical jargon.
- Use **headings, subheadings, bullet points, and visuals** to make the report easy to read.
- Support text with **tables, charts, and figures** whenever possible.
- Always **cite sources** (APA/IEEE).
- **Proofread** carefully for grammar, clarity, and consistency.

!Think of your report as **telling the story of your engineering project journey**: What problem you faced, how you tackled it, what you found, and why it matters.

4. Research Ethics (Plagiarism, Citation & Referencing)

4.1. Plagiarism (What to Avoid):

- **Copying:** Using the same or very similar words from a source without quotation marks or citation.
- **Paraphrasing poorly:** Changing only a few words but keeping the original sentence structure.
- **Collusion:** Working with others but submitting as if it were only your work.
- **Duplication:** Submitting the same work for different courses.

Golden Rules:

- Always acknowledge sources.
- Use your own academic voice.
- Be transparent so others can reproduce your work.

4.2. Citation vs Referencing:

- **Citation (in-text):** A short signal in your text showing the source.
 - *APA:* (Doe, 2021)
 - *IEEE:* [1]
- **Reference (end list):** Full details so readers can find the source.
 - *APA:* Doe, J. (2021). *Urban Wind Effects*. City Press.
 - *IEEE:* [1] J. Doe, *Urban Wind Effects*. City Press, 2021.

Key point: Citation = brief, in-text; Reference = full, at the end.

4.3. Direct Quotation vs Paraphrase

- **Original:**
“Wind speeds frequently intensify at the windward corners of high-rise buildings due to flow constriction and downward deflection along the façade.”
- **Direct Quote (word-for-word):**
“Wind speeds frequently intensify at the windward corners of high-rise buildings due to flow constriction and downward deflection along the façade.” (Doe, 2021, p. 45)
- **Good Paraphrase (own words, same meaning):**
Wind velocity often increases near the upwind edges of tall buildings because the façade funnels airflow downward toward pedestrians (Doe, 2021).
- **Bad Paraphrase (too close to original):**
Wind speeds commonly rise at the windward corners of high-rise buildings because of flow restriction and downward deflection (Doe, 2021).

Tip:

Only **5–10%** of your report should be direct quotes. Do not make a direct quote unless necessary. Most of it should be paraphrased properly with citations.

4.4. Research Ethics Checklist

Before submitting your report, ask yourself:

- **Sources**
 - Did I cite every idea, figure, or sentence that is not my own?
 - Did I include full references at the end for every in-text citation?
- **Paraphrasing & Quoting**
 - Did I rewrite information in my **own words and structure**?
 - Are direct quotes limited to only essential parts (max 5–10%)?
 - Did I avoid “bad paraphrasing” (just swapping a few words)?
- **Originality & Integrity**
 - Did I avoid copying text without acknowledgement?
 - Did I make sure the work is genuinely ours as a team (no collusion)?
 - Am I avoiding recycling old assignments (no duplication)?

If you can answer “**Yes**” to all, your report is likely safe from plagiarism issues.

5. Teamwork

Successful project work in ENG 401 depends not only on technical skills but also on **effective teamwork**.

- **Assign tasks according to expertise.** Fig. 5 illustrates how task allocation can be made between Geomatics Engineering (GE) and Mechanical Engineering (ME): GE focusing on spatial data analysis, such as determining building height/coordinates and creating 3D models or site plans, while ME focuses on simulation tasks, such as aerodynamic evaluation.
- **Maintain clear communication channels.** Use shared platforms (Teams, Slack) to track updates.

- **Plan time carefully.** Set milestones, track version history, and maintain accountability. (Fig. 6 shows an example time schedule using a simple Gantt chart.)
- **Think critically.** Continuously question assumptions, test results, and evaluate limitations.
- **Share drafts early.** Circulate reports among team members and supervisors for constructive feedback.
- **Use visual tools.** Task/time charts and responsibility matrices provide clarity and prevent overlapping responsibilities.

! By clearly assigning tasks and task/time charts according to each discipline's expertise, teams can work efficiently, reduce errors, and leverage each member's strengths. Understanding your role not only helps in project success but also builds practical experience in interdisciplinary collaboration, which is vital for real-world engineering projects.

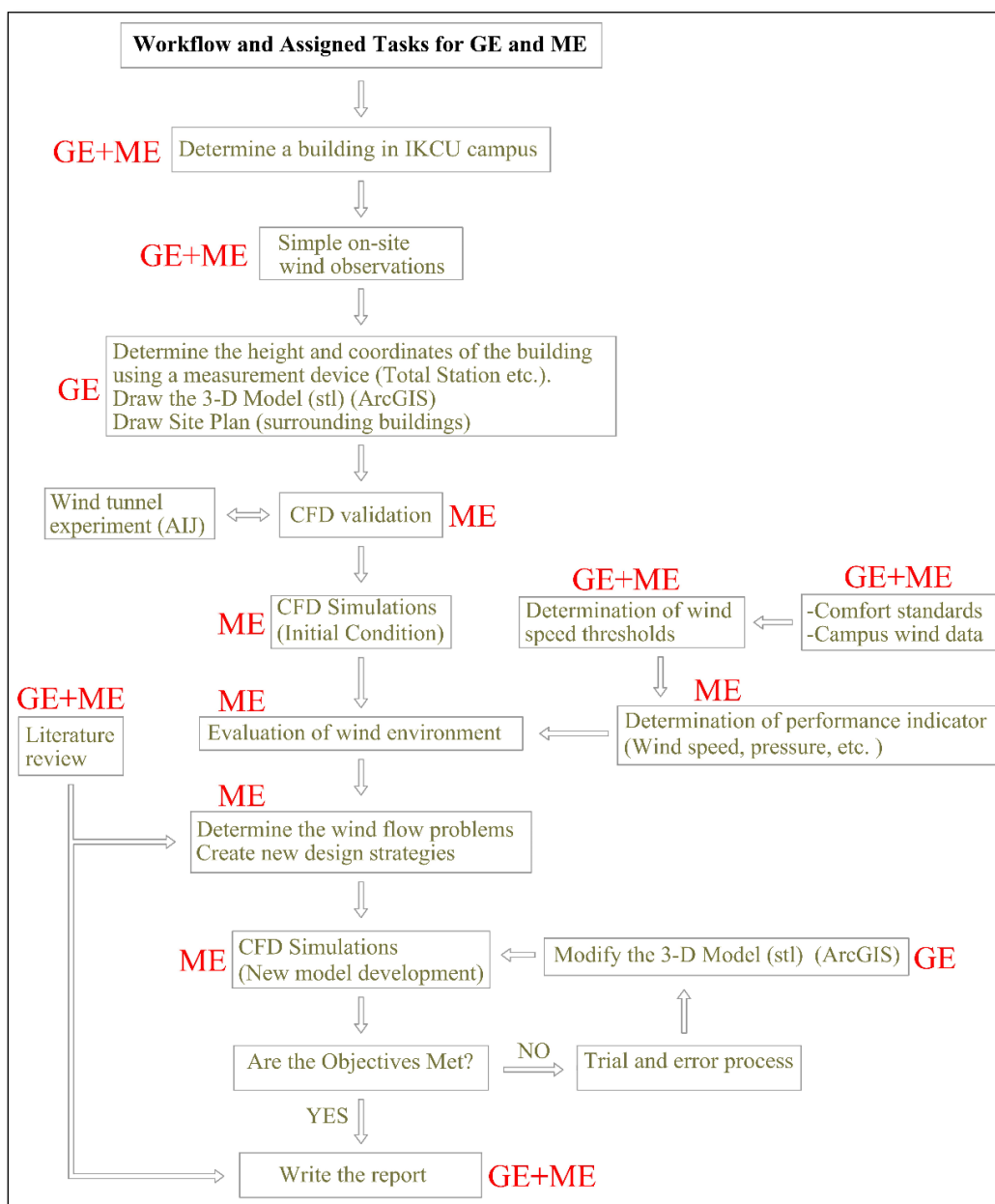


Fig. 5 Workflow and assigned tasks according to expertise.

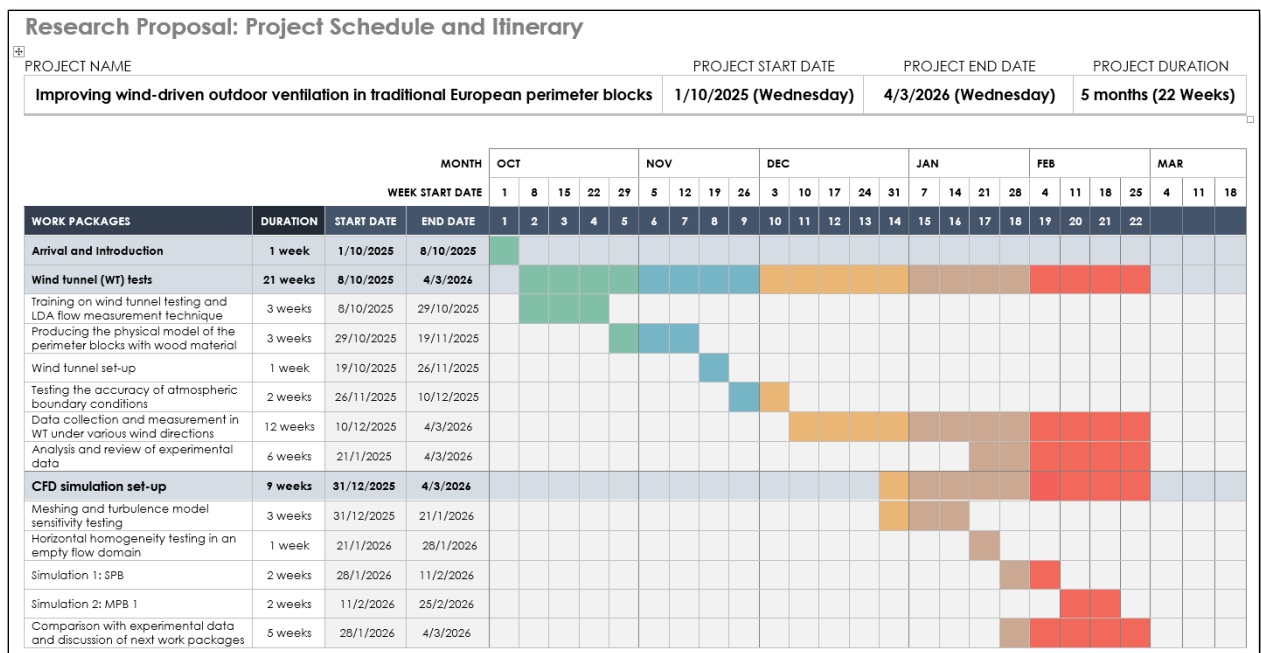


Fig 6. Task/time charts.

Conclusion

Successfully completing an engineering project requires a combination of careful research, ethical reporting, and effective teamwork. **Identifying a clear research problem, reviewing relevant literature, selecting appropriate methods, and documenting findings rigorously** ensures that your work is **reliable** and **reproducible**.

Writing **structured, clear, and well-referenced reports** allows you to communicate your design journey effectively. Moreover, distributing tasks according to each team member's expertise and maintaining open communication helps the team work efficiently and achieve better solutions. Mastering these skills not only leads to successful projects but also prepares you for real-world interdisciplinary engineering challenges.